

Seokyung Park

M.S. Student · Agricultural Robotics and Smart Sensing

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Research interests

Agricultural robotics, mobile environmental sensing, crop phenotyping and physical measurement systems. My research combines experimental measurements, image analysis and numerical modelling to study crops and agricultural environments.

Education

Pusan National University

Mar. 2025 – Feb. 2027 (expected)

M.S. in Bio-Industrial Machinery Engineering · GPA: 4.36/4.50

Precision Agriculture and Smart Sensing System Lab.

Pusan National University

Mar. 2017 – Feb. 2025

B.S. in Plant Bioscience · GPA: 4.26/4.50

Summa Cum Laude; President's List.

Research experience

Graduate Researcher

Mar. 2025 – Present

Precision Agriculture and Smart Sensing System Lab., Pusan National University

Damage-aware robotic gripping for tomato harvesting

- Investigated loading-rate effects on compression damage in standard-sized and cherry tomatoes using a myCobot 320 Pi arm, adaptive gripper and force sensor.
- Combined force–deformation analysis with hyperspectral measurements at 1 hour and 5 days after compression to study delayed damage; contributed to experimental design, analysis and manuscript preparation as first author.

Quadruped robotics for agricultural sensing

- Contributed to greenhouse microclimate mapping with a Unitree Go2 platform, measuring temperature, relative humidity and CO₂ at two canopy heights across eight lanes. Co-first author of the greenhouse manuscript.
- Worked on reservoir upstream surveys using RGB imaging and RTK-GNSS to construct location-linked ground-image datasets. Plant phenotyping is a planned extension of the sensing platform.

Uplift resistance of expandable foundation piles

- Used Abaqus/Standard finite-element models to examine how wing inclination and dimensions influence pile uplift resistance in sandy soil; analysed load–displacement behaviour and soil plastic zones.

Load-cell sensing for air-supported domes

- Participated in load-cell installation and field acquisition at Damyang on 15 June 2026, recording cable-tension data through an indicator, USB-to-RS485 interface and Raspberry Pi.

Undergraduate Research Student

Jul. 2023 – Feb. 2024

Functional Crop Laboratory, Pusan National University

- Worked on soybean leaf phenotyping under drought using time-series RGB images, YOLOv8, Roboflow and ImageJ, with Arduino-based environmental measurements.

Publications and manuscripts

Peer-reviewed publication

- [1] J. Park, S. Park, E. Jung, J. Song, G. Park, J. Lee, M. Kim, and J. Park. Inertial sensor applications in agriculture: Sensor types and data utilization techniques for mechanical and wearable applications. *Korean Journal of Agricultural Science*, 52(4):447–469, 2025. doi: 10.7744/kjoas.520405.

Under review

Park, S., Song, J., Hwang, H., Park, G., and Park, J. (2026). Speed-dependent control of robotic gripper for tomato harvesting to minimise compression-induced damage. *Biosystems Engineering*. **Under review, second revision.**

Manuscripts in final preparation before submission

Hwang, H.[†], **Park, S.**[†], and Park, J. Greenhouse microclimate mapping using a quadruped robot to assess fixed sensor representativeness. [†]Equal contribution.

Park, S., Park, J., Jang, J., Park, D., Kwon, S., Kim, M., and Choi, G. Analysis of uplift resistance characteristics of expandable piles in sandy soil with varying wing inclination and dimensions. *English translation of the Korean manuscript title*.

Research funding

Individual research fellowship

Master's Student Research Encouragement Grant

Sep. 2025 – Aug. 2026

Science and Engineering

Development of Locomotion Algorithms and Autonomous Control Systems for Quadruped Robots Specialized for Korean Agricultural Terrain.

Laboratory project support

- **Quadruped robotics:** National Research Foundation of Korea (NRF), funded by the Ministry of Science and ICT (MSIT). Grant **RS-2026-25475333**.
- **Air-dome foundations and sensing:** Korea Institute of Planning and Evaluation for Technology in Food, Agriculture and Forestry (IPET), through the Agri-Food Export Enhancement Technology Development Program, funded by the Ministry of Agriculture, Food and Rural Affairs (MAFRA). Grant **RS-2025-02313370**.

Technical skills

- **Programming and analysis:** Python, NumPy, pandas and SciPy; force-signal processing, data visualisation and model fitting.
- **Robotics and acquisition:** ROS 2, Unitree Go2, Raspberry Pi, Arduino, load cells, force sensors, RTK-GNSS, IMU and RS485 interfaces.
- **Imaging:** RGB and hyperspectral image analysis; YOLOv8, Roboflow and ImageJ.
- **Numerical modelling:** Abaqus/CAE and Abaqus/Standard; finite-element analysis and contact modelling.

Selected conference presentations

- **Park, S.** (14 July 2026). Height-resolved 3D microclimate mapping and spatiotemporal analysis within greenhouse crop canopies using a quadruped robot. *AKABFE Student Research Spotlight, ASABE Annual International Meeting*, Indianapolis, USA. Presenter.
- **Park, S.,** Park, J., Hwang, H., and Kang, S. (2026). Ground-reaction-force-based lateral slope estimation for furrow-following quadruped robots. *Korean Society for Agricultural Machinery Spring Conference*.
- **Park, S.,** Park, J., Song, J., and Kim, M. (31 October 2025). Analysis of pile pull-out behaviour in Middle Eastern soil using digital image correlation. *Korean Society of Agricultural Engineers Autumn Conference*, Jeju.
- Hwang, H., **Park, S.,** et al. (22 May 2025). Remote control and integrated sensor-data environment for an agricultural quadruped robot. *Korean Society for Precision Agriculture Spring Conference*. Co-author.
- **Park, S.,** Park, J., Park, G., Song, J., and Hwang, H. (25 April 2025). Minimising damage and optimising control in fruit and vegetable harvesting using a robotic arm and gripper with real-time force sensing. *Korean Society for Agricultural Machinery Spring Conference*.

English translations are used for Korean presentation titles.

Honours and awards

- **Best Paper Presentation Award (team)**, Korean Society for Precision Agriculture, May 2025 and November 2024.
- **Excellent Paper Presentation Award**, Korean Society for Agricultural Machinery, April 2025 and October 2024.
- **Excellence Award**, PNU–PKNU Field Problem Research Group Final Symposium, November 2024.
- **Best Presentation Award (team)**, Korean Society for Engineering Education, September 2024.
- **Bronze Prize**, PNU College of Natural Resources and Life Science Capstone Design Competition, October 2023.
- **Excellence Award**, Busan LINC 3.0 Entrepreneurship Shared University Competition, January 2023.
- **PNU merit scholarships**: four full-tuition and two partial-tuition awards, 2017–2024.

Selected coursework

Graduate: Biological Image Processing; Statistical Processing and Data Analysis in Bio-Industrial Machinery Engineering; Advanced Bio-Industrial Machinery Design; Biosystems Mechanics.

Undergraduate: Biosensor and Bio-Instrumentation and Practice; Biosystems Control Engineering; Bio-Production Mechatronics Design; Experimental Design for Biologists; Plant Physiology and Practice; Plant Genetics.

Additional training

Modeling Contact with Abaqus/Standard, VPK PLM Solution Training Center, 25–26 June 2026.

Introduction to Abaqus/CAE, VPK PLM Solution Training Center, 18–20 March 2026.